Influences of terrain on snow depth at very high resolution

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EARTH OBSERVATION RESEARCH

1. Abstract

UAV LiDAR data was used to derive digital surface models (DSMs) of the area of interest at eight separate occasions from 2024 to 2026. UAV survey flights were conducted between September and April. A snowfree DSM was then used to calculate snow depth as well as slope, northness, wind shelter index (WSI) and the topographic position index (TPI) at a resolution of 10 cm. Using a Generalized Additive Model (GAM), the terrain parameters were used to explain the mean snow depth. The model performed with an R^2 of 0.46.

2. Introduction

Snow serves a number of ecological functions and has a strong influence of soil temperature, water discharge and local albedo. On the south face of Zugspitze, snow cover and depth is also of economic interest for the ski resort. The influence of terrain conditions on snow depth is well known and terrain derived indices are often used in modelling of snow depth [1-5]. This paper will focus on the influence of terrain on snow depth at very high resolution.

3. Study area

The 0.23 km² study area is located on the south face of the Zugspitze in the German Alps as shown in Fig. 1. Ranging from 2385 to 2568 m a.s.l., the site consists primarily of moraine debris and limestone.

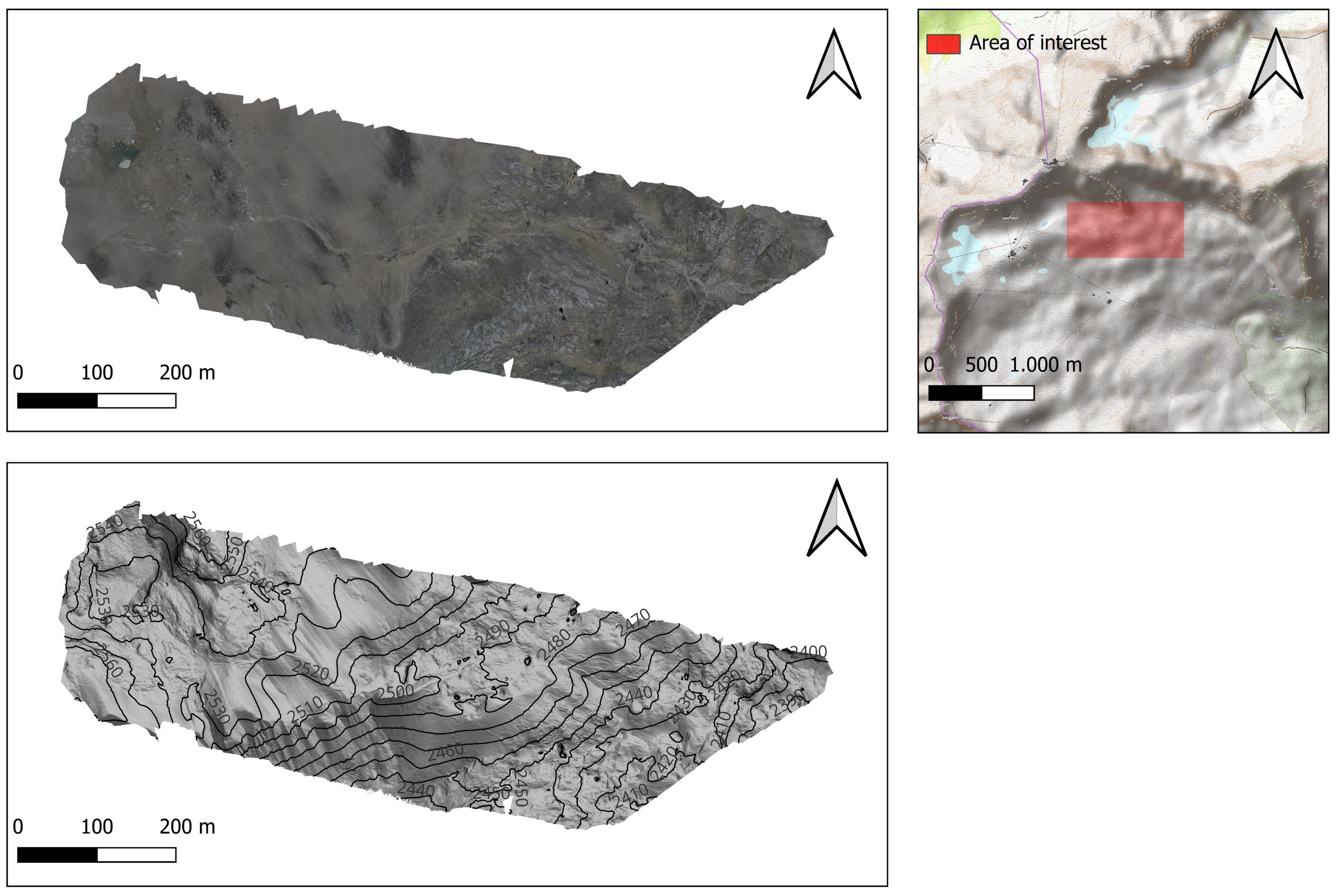

Fig. 1.: Map of the area of interest

4. Methods

4.1 Snow depth calculation

LiDAR data were acquired using a DJI Zenmuse L1 sensor at the resolution of 10cm. A snowfree DSM was calculated using the smallest values for each pixel across all DSMs. Snow depth was calculated as the difference between the snow-on and snowfree DSMs. To derive the mean snow depth, all snow depth models were standardized, and the mean of them was calculated. The data was compared with in-situ measured snow depth. The snow depth was measured manually with metal rods. To process the geospatial data, the terra-package was used [6].

4.2 Relief parameter calculation

Additionally, the snowfree DSM was used to calculate the relief indices in SAGA GIS [7]. Slope and northness were calculated to capture the impact of the incoming radiation. The TPI was calculated to account for the tendency of snow to aggregate in concave positions [8]. The optimal radius of 28 meters for the TPI was determined using the tpiscaleR-package [9]. The WSI was used to reflect snow deposition by wind [2]. The therefore necessary mean wind direction was derived from the ERA5 dataset using Google Earth Engine for 2020 to 2026 [10]. The WSI was calculated with a range of 100 m and a tolerance of 30°.

4.3 Generalized Additive Model

The GAM was trained on a basis of 2.000 randomly sampled pixels. 500 pixels were used for cross validation. Because the effects of northness and slope interact, a tensor product was used on both of them. TPI, elevation and WSI was modelled using a smooth terms. The modelling was done using the mgcv-package [11].

5. Results

5.1 Snow depth calculation accuracy

The LiDAR-based snow depth estimation shows high agreement with the in-situ measurements. For the higher values of the LiDAR-based values, the in-situ measured snow depth is significantly lower as shown in Fig. 2.

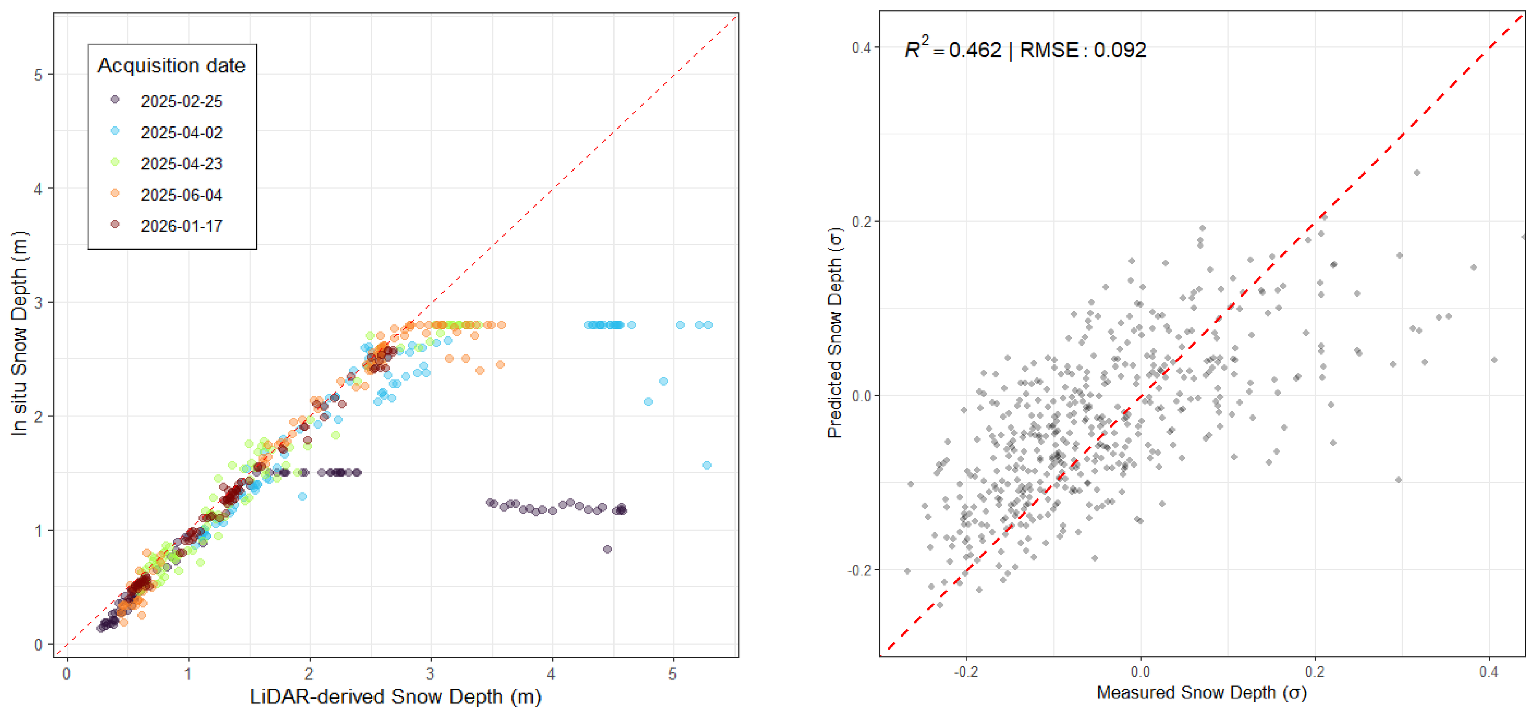

Fig. 2: Scatterplot of the LiDAR- and in-situ snowdepth estimation, Scatterplot of GAM

5. 2 GAM performance and feature effects

The GAM performed with an out-of-sample R^2 of 0.46 and a RMSE of 0.092 σ as shown in Fig. 2. All of the predictors scored p-values lower than $2{\cdot}10^{-16}$ and can therefore be seen as highly significant (Fig. 3). Elevation and TPI both have significantly higher F-statistics than WSI and the combination of slope and northness.

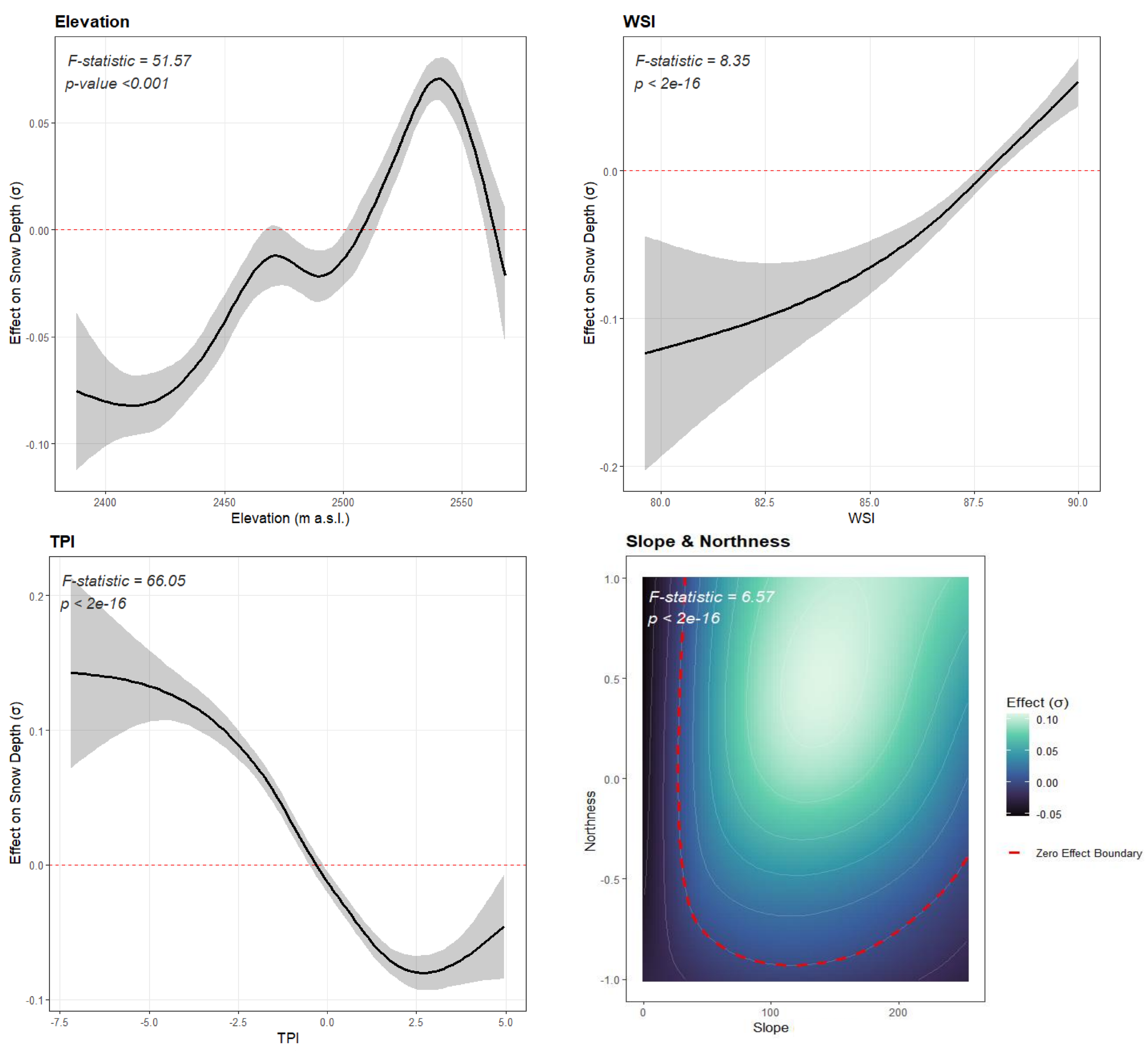

Fig. 3: Smooth terms of the variables used in the GAM

6. Discussion and Conclusion

The differences between in-situ measured snow depth and LiDAR-based snow depth can be attributed to the measurement using rods.
Discrepancies occur when manual rods strike hard firn layers or a different obstacle at greater depth.

The smooth terms suggest the increasing snow depth with higher WSI and lower TPI values. This effect can be explained with the aeolian redistribution of snow. The model also suggests higher snow depth with higher elevation which can be attributed to lower temperatures in higher elevations. However, snow depth exhibits a decline at peak elevations. This might be linked to high TPI values in these elevated areas and the dominant effect of TPI overall. The compound effects of slope and northness can be attributed to the incoming radiation as well as the limited accumulation of snow on steep surfaces.
The GAM performance is in the range of comparable studies that use linear models [3,4].

Sources can be viewed using the QR-Code in the upper-left corner.